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JT/ T 1078 -2016

GNSS System for operating vehicles-  
video communication protocol

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## Preface

This standard is drafted in accordance with the rules given in GB/T 1.1-2009 given rules for drafting.

This standard is proposed by the Technical Committee for Standardization of Transportation Information Communication and Navigation and is under the control of the Technical Committee.

Drafting unit of this standard: China Communications and Information Center, Beijing  
Guojiao Xintong Technology Development Co., Ltd, Jiangsu Weisen Communication Technology Co., Ltd, Dalian Xinkai Digital Software Co. Ltd., Hangzhou Hikvision Digital Technology Co.

Main drafters of this standard: Feng Quan, Liu Jian, Niu Wenjiang, Wang Shufang, Wang Quan, Yang Shengjun, Dong Hongjun, Zeng Zhuo, Chen Weiwei, Yu Zhikai, Liu Tianying, Li Tao, Zhang Zeqi, Lin Yuan.

## Satellite positioning system for road transport vehicles video communication protocol

### 1 Scope

This standard specifies the road transportation vehicle satellite positioning system, vehicle video terminal and video platform between the basis of the agreement and communication protocols, audio and video streaming server and the client playback software between the code streaming communication, as well as the video platform between the basis of the communication protocol, communication protocol flow, constant definitions and the agreement data format.

This standard applies to the transmission of audio and video data between the on-board video terminal of the satellite positioning system of road transportation vehicles and the enterprise video monitoring platform, as well as the exchange and sharing of audio and video resources between different video platforms.

### 2 Normative references

The following documents are essential to the application of this document . Where a document is cited with a date, only the dated version applies to this document. The latest version of the document (including all change orders) applies to this document. Where a document is cited without date, the latest version (including all change orders) applies to this document.

JT/T 808-2011 Road Transportation Vehicle Satellite Positioning System Terminal Communication Protocol and Data Format

JT/T 809-2011 Satellite Positioning System for Road Transportation Vehicles Platform Data Exchange

JT/T 1076-2016 Technical Requirements for On-board Video Terminal of Road Transportation Vehicle Satellite Positioning System

JT/T 415-2006 Rules for Cataloging and Coding of Road Transportation e-Government Platforms

IETF RFC 3550 RTP Real-time Transport Protocol (RTP)

IETF RFC 2854 The Text/Html Media Type for Text/Hypertext Markup Language

### 3 Terms and definitions, acronyms

#### 3.1 Terms and definitions

The following terms and definitions apply to this document.

##### 3.1.1

code rate

The number of bits of data transmitted per unit of time during data transmission, commonly used in kilobits per second (kbps).

### 3.1.2

#### Frame rate

Indicates the number of times per second that the graphics processor is able to update a field while processing it, and is used as a measure of the number of frames displayed, measured in Frames per Second (FPS).

### 3.2 Abbreviations

The following abbreviations apply to this document.

AAC: Advanced Audio Coding

MPEG: Moving Pictures Experts Group

RTP: Real-time Transport Protocol

TCP: Transmission Control Protocol

UDP: User Datagram Protocol

URL: Uniform Resource Locator ( U R L )

UTF-8: Unicode (8-bit Unicode Transformation Format)

FTP: File Transfer Protocol (FTP)

#### 4 Foundation of protocols between video terminals and video platforms

##### 4.1 Basic agreement

The communication mode, data type, transmission rule and message composition of the protocol are in accordance with the requirements of Chapter 4 of JT/T 808-2011.

The communication connection mode of signaling data message in the protocol is in accordance with the requirements of Chapter 5 of JT/T 808-2011.

The message processing mechanism of signaling data message in the protocol is in accordance with the requirements of Chapter 6 of JT/T 808-2011.

The encryption mechanism of signaling data message in the protocol is in accordance with the requirements of Chapter 7 of JT/T 808-2011.

The agreement shall meet the following requirements for all parties to platform and terminal communications:

- All messages should be answered unless expressly agreed;
- A generic response shall be used for responses that do not explicitly specify a dedicated response message;
- For messages with sub-packages, the responder shall respond to each sub-package on a packet-by-packet basis.

##### 4.2 Real-time audio and video transmission channel conventions

A real-time audio/video transmission channel can transmit one video message or one audio message, or one video message and one audio message. There are two types of real-time audio/video transmission channel conventions as follows:

-When TCP is used, each TCP connection can carry multiple audio and video channels. If there is no number of channels within the set timeout period, the TCP connection can carry multiple channels.

Both the terminal and the monitoring center can actively close the TCP connection used for audio and video data transmission. -When UDP is used, each UDP port can carry multiple audio and video channels.

##### 4.3 Classification of audio and video communication messages

Audio and video data messages are divided into the following two categories:

- Signaling data message: the data format should conform to the provisions of JT/T 808 -2011, and new protocol commands and data format should be added on the basis of its protocol format. The message communication should use the link already established between the vehicle-mounted video terminal and the



enterprise video monitoring platform for transmitting positioning information, and no new link should be established.

- Code stream data message: used for network real-time audio and video transmission, network video playback, voice dialog, voice monitoring, voice broadcasting, etc. The message communication should be on a new link without using the link that transmits the location information.

## 5 Communication protocols between video terminals and video platforms

### 5.1 Protocol Instruction Set

For the comparison table of command messages between video terminals and video platforms, see Appendix A

### 5.2 Inheritance Directives

Inherit other commands in JT/T 808-2011 except message ID 0x8804 (recording start command). In addition, the multimedia type field in the five commands 0x0800 (multimedia event message upload), 0x0801 (multimedia data upload), 0x8802 (store multimedia data retrieval), 0x0802 (store multimedia data retrieval response), 0x8803 (store multimedia data upload) in JT/T 808-2011 shall contain only the picture type in this standard. In this standard, only the picture type shall be included, and the data of audio and video type shall be in accordance with 5.4 and 5.5. Audio and video type data shall be transferred according to the requirements of 5.4 and 5.5. Audio and video data shall be transmitted according to the requirements of 5.4 and 5.5.

### 5.3 Parameter setting commands

#### 5.3.1 Terminal audio and video parameter settings

The terminal audio/video parameter setting message adopts the 0x8103 message defined in 8.8 of JT/T 808-2011 and adds the following audio/video parameter settings, see Table 1. 8 of JT/T 808-2011, and add the following audio and video parameter settings, see Table 1.

Table 1 Audio and video setting parameter table

| Parameter ID | Data Type | Description and Requirements                                                                                                                                                  |
|--------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x0075       |           | Audio and video parameter settings, see Table 2 for description                                                                                                               |
| 0x0076       |           | Audio and video channel list settings, see Table 3 for description                                                                                                            |
| 0x0077       |           | Individual video channel parameter settings, described in Table 5                                                                                                             |
| 0x0079       |           | Special alarm recording parameter settings, see Table 7 for description                                                                                                       |
| 0x007A       | DWORD     | Video related alarm mask word, corresponds to the definition of video alarm flag bit in Table 13, the corresponding bit is 1 then the corresponding type of alarm is blocked. |
| 0x007B       |           | Image analysis alarm parameter settings, described in Table 8                                                                                                                 |
| 0x007C       |           | Terminal sleep-wake mode settings, see Table 9 for descriptions.                                                                                                              |

Table 2 Definitions and descriptions of audio and video parameters

| Starting Byte | field                          | data type | Description and Instructions                                                                                     |
|---------------|--------------------------------|-----------|------------------------------------------------------------------------------------------------------------------|
| 0             | Live Streaming Encoding Mode   | BYTE      | 0: CBR (fixed code rate);<br>1: VBR (Variable Bit Rate);<br>2: ABR (Average Bit Rate);<br>100 to 127: Customized |
| 1             | Real-time streaming resolution | BYTE      | 0: QCIF;<br>1: CIF;<br>2: WCIF;<br>3: D1;<br>4: WD1;<br>5: 720p;<br>6: 1080P;<br>100 to 127: Customized          |

|   |                                  |       |                                                                                                                  |
|---|----------------------------------|-------|------------------------------------------------------------------------------------------------------------------|
| 2 | Live Streaming Keyframe Interval | WORD  | Range (1 to 1,000) frames                                                                                        |
| 4 | Live streaming target frame rate | BYTE  | Range (1 to 120) frames/s                                                                                        |
| 5 | Live Streaming Target Bitrate    | DWORD | In kilobits per second (kbps)                                                                                    |
| 9 | Streaming Encoding Mode          | BYTE  | 0: CBR (fixed code rate);<br>1: VBR (Variable Bit Rate);<br>2: ABR (Average Bit Rate);<br>100 to 127: Customized |

Table 2 (continued)

| Starting Byte | field                            | data type | Description and Instructions                                                                                                                                                                                                                                                                                                             |
|---------------|----------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10            | Stored Stream Resolution         | BYTE      | 0: QCIF;<br>1: CIF;<br>2: WCIF;<br>3: D1;<br>4: WD1;<br>5: 720p;<br>6: 1080P;<br>100 to 127: Customized                                                                                                                                                                                                                                  |
| 11            | Memory Stream Keyframe Interval  | WORD      | Range (1 to 1,000) frames                                                                                                                                                                                                                                                                                                                |
| 13            | Storage stream target frame rate | BYTE      | Range (1 to 120) frames/s                                                                                                                                                                                                                                                                                                                |
| 14            | Storage Stream Target Bitrate    | DWORD     | In kilobits per second (kbps)                                                                                                                                                                                                                                                                                                            |
| 18            | OSD subtitle overlay settings    | WORD      | Set by bit: 0 for no stacking, 1 for stacking;<br>bit0: date and time;<br>bit1: License plate number;<br>bit2: logical channel number;<br>bit3: latitude and longitude;<br>bit4: Traveling record speed;<br>bit5: Satellite positioning speed;<br>bit6: continuous driving time;<br>bit7 ~ bit10: Reserved;<br>bit11 ~ bit15: Customized |
| 20            | Whether to enable audio output   | BYTE      | 0: not enabled; 1: enabled                                                                                                                                                                                                                                                                                                               |

Table 3 Audio and video channel list

| Starting Byte | field                                    | data type                 | Description and Instructions |
|---------------|------------------------------------------|---------------------------|------------------------------|
| 0             | Total number of audio and video channels | BYTE                      | denoted by l                 |
| 1             | Total number of audio channels           | BYTE                      | Denote by m                  |
| 2             | Total number of video channels           | BYTE                      | denoted by n                 |
| 3             | Audio/Video Channel Comparison Table     | BYTE<br>[4 × (l + m + n)] | See table 4                  |

Table 4 Audio/Video Channel Comparison Table

| Starting Byte | field                   | data type | Description and Instructions           |
|---------------|-------------------------|-----------|----------------------------------------|
| 0             | Physical Channel Number | BYTE      | Starting at 1                          |
| 1             | logical channel number  | BYTE      | According to Table 2 in JT/T 1076-2016 |

Table 4 (continued)

| Starting Byte | field                       | data type | Description and Instructions                                                  |
|---------------|-----------------------------|-----------|-------------------------------------------------------------------------------|
| 2             | Channel Type                | BYTE      | 0: Audio and video;<br>1: Audio;<br>2: Video                                  |
| 3             | Whether to connect the head | BYTE      | This field is valid for channel types 0 and 2; 0: not connected; 1: connected |

Table 5 Individual channel video parameter definitions and descriptions

| Starting Byte | field                                                                     | data type    | Description and Instructions |
|---------------|---------------------------------------------------------------------------|--------------|------------------------------|
| 0             | Number of channels for which video parameters need to be set individually | BYTE         | denoted by n                 |
| 1             | Individual Channel Video Parameter Setting List                           | BYTE[21 × n] | See table 6                  |

Table 6 Individual Channel Video Parameter Settings

| Starting Byte | field                            | data type | Description and Instructions                                                                                     |
|---------------|----------------------------------|-----------|------------------------------------------------------------------------------------------------------------------|
| 0             | logical channel number           | BYTE      | According to Table 2 in JT/T 1076-2016                                                                           |
| 1             | Live Streaming Encoding Mode     | BYTE      | 0: CBR (fixed code rate);<br>1: VBR (Variable Bit Rate);<br>2: ABR (Average Bit Rate);<br>100 to 127: Customized |
| 2             | Real-time streaming resolution   | BYTE      | 0: QCIF;<br>1: CIF;<br>2: WCIF;<br>3: D1;<br>4: WD1;<br>5: 720p;<br>6: 1080P;<br>100 to 127: Customized          |
| 3             | Live Streaming Keyframe Interval | WORD      | Range (1 to 1,000) frames                                                                                        |
| 5             | Live streaming target frame rate | BYTE      | Range (1 to 120) frames/s                                                                                        |
| 6             | Live Streaming Target Bitrate    | DWORD     | In kilobits per second (kbps)                                                                                    |

|    |                         |      |                                                                                                                  |
|----|-------------------------|------|------------------------------------------------------------------------------------------------------------------|
| 10 | Streaming Encoding Mode | BYTE | 0: CBR (fixed code rate);<br>1: VBR (Variable Bit Rate);<br>2: ABR (Average Bit Rate);<br>100 to 127: Customized |
|----|-------------------------|------|------------------------------------------------------------------------------------------------------------------|

Table 6 (continued)

| Starting Byte | field                            | data type | Description and Instructions                                                                                                                                                                                                                                                                                                                 |
|---------------|----------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11            | Stored Stream Resolution         | BYTE      | 0: QCIF;<br>1: CIF;<br>2: WCIF;<br>3: D1;<br>4: WD1;<br>5: 720p;<br>6: 1080P;<br>100 to 127: Customized                                                                                                                                                                                                                                      |
| 12            | Memory Stream Keyframe Interval  | WORD      | Range (1 to 1,000) frames                                                                                                                                                                                                                                                                                                                    |
| 14            | Storage stream target frame rate | BYTE      | Range (1 to 120) frames/s                                                                                                                                                                                                                                                                                                                    |
| 15            | Storage Stream Target Bitrate    | DWORD     | In kilobits per second (kbps)                                                                                                                                                                                                                                                                                                                |
| 19            | OSD Overlay Settings             | WORD      | Set by bit: 0 means no overlay, 1 means overlay;<br>bit0: date and time;<br>bit1: License plate number;<br>bit2: logical channel number;<br>bit3: latitude and longitude;<br>bit4: Traveling recorded speed;<br>bit5: Satellite positioning speed;<br>bit6: continuous driving time;<br>bit7 ~ bit10: Reserved;<br>bit11 ~ bit15: Customized |

Table 7 Special Alarm Video Parameter Definitions and Descriptions

| Starting Byte | field                                  | data type | Description and Instructions                                                                                                   |
|---------------|----------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------|
| 0             | Special alarm video storage thresholds | BYTE      | Special alarm video occupies the main memory storage threshold percentage, take the value of 1 to 99, the default value is 20. |
| 1             | Special Alarm Video Duration           | BYTE      | Maximum duration of special alarm recording in minutes (min), default value 5                                                  |
| 2             | Special alarm marking start time       | BYTE      | The recording time in minutes before the special alarm occurs, the default value is 1.                                         |

Table 8 Video analysis alarm parameter definitions and descriptions

| start byte | field | data type | Description and Instructions |
|------------|-------|-----------|------------------------------|
|------------|-------|-----------|------------------------------|



|   |                                                         |      |                                                                                                             |
|---|---------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------|
| 0 | Number of persons authorized to be carried in a vehicle | BYTE | Authorized number of passengers on passenger vehicles, video analysis results Alarm generated when exceeded |
| 1 | Fatigue threshold                                       | BYTE | Video analyzes fatigue driving alarm thresholds and generates an alarm when exceeded                        |

Table 9 Terminal sleep-wake mode setting data format

| Starting Byte | field                           | data type | Description and Requirements                                                                                                                                                                                                        |
|---------------|---------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0             | Sleep Wake Mode                 | BYTE      | Setting by bit: 0 means no setting, 1 means setting;<br>bit0: Conditional wakeup;<br>bit1: Timed wakeup;<br>bit2: manual wakeup                                                                                                     |
| 1             | Wake-up condition type          | BYTE      | This field is valid when bit0 is 1 in sleep-wake mode, otherwise it is set to 0;<br>Setting by bit: 0 means no setting, 1 means setting;<br>bit0: Emergency alarm;<br>bit1: Collision rollover alarm;<br>bit2: Vehicle door opening |
| 2             | Timed Wake-Up Day Setting       | BYTE      | Setting by bit: 0 means no setting, 1 means setting;<br>bit0: Monday;<br>bit1: Tuesday;<br>bit2: Wednesday;<br>bit3: Thursday;<br>bit4: Friday;<br>bit5: Saturday;<br>bit6: Sunday                                                  |
| 3             | Day Timer Wakeup Parameter List | BYTE[17]  | See table 10, time periods should not overlap                                                                                                                                                                                       |

Table 10 Definitions of Day Wakeup Parameters

| Starting Byte | field                      | data type | Description and Requirements                                                                                                                                                                                                             |
|---------------|----------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0             | Timed Wakeup Enable Flag   | BYTE      | Setting by bit: 0 means no setting, 1 means setting;<br>bit0: Time period 1 wake up time enabled;<br>bit1: time period 2 wake up time enabled;<br>bit2: Time period 3 Wake - up time enabled;<br>bit3: time period 4 wake up time enable |
| 1             | Time period 1 Wake-up time | BCD[2]    | HHMM, value range 00:00 ~ 23:59                                                                                                                                                                                                          |
| 3             | Time period 1 Closing time | BCD[2]    | HHMM, value range 00:00 ~ 23:59                                                                                                                                                                                                          |
| 5             | Time period 2 Wake-up time | BCD[2]    | HHMM, value range 00:00 ~ 23:59                                                                                                                                                                                                          |
| 7             | Time period 2 Closing time | BCD[2]    | HHMM, value range 00:00 ~ 23:59                                                                                                                                                                                                          |

|    |                            |        |                                 |
|----|----------------------------|--------|---------------------------------|
| 9  | Time period 3 Wake-up time | BCD[2] | HHMM, value range 00:00 ~ 23:59 |
| 11 | Time period 3 Closing time | BCD[2] | HHMM, value range 00:00 ~ 23:59 |
| 13 | Time period 4 Wake-up time | BCD[2] | HHMM, value range 00:00 ~ 23:59 |
| 15 | Time period 4 Closing time | BCD[2] | HHMM, value range 00:00 ~ 23:59 |

## 5.3.2 Querying terminal audio and video attributes

Message ID: 0x9003.

The message body is empty.

## 5.3.3 Terminal upload audio and video attributes

Message ID: 0x1003.

Message type: signaling data message.

The terminal uploading audio/video attribute command is used to respond to the query terminal audio/video attribute message sent by the platform, and the data format of the message body is shown in Table 11.

Table 11 Data format of audio and video attributes uploaded from terminals

| Starting Byte | field                                                                  | data type | Description and Requirements                    |
|---------------|------------------------------------------------------------------------|-----------|-------------------------------------------------|
| 0             | Input audio encoding method                                            | BYTE      | See table 12                                    |
| 1             | Number of input audio channels                                         | BYTE      |                                                 |
| 2             | Input Audio Sample Rate                                                | BYTE      | 0:8kHz;<br>1:22.05kHz;<br>2:44.1kHz;<br>3:48kHz |
| 3             | Input Audio Sample Bits                                                | BYTE      | 0:8bits;<br>1:16bits;<br>2:32-bit               |
| 4             | Audio Frame Length                                                     | WORD      | Range 1 to 4 294 967 295                        |
| 6             | Whether to support audio output                                        | BYTE      | 0: not supported; 1: supported                  |
| 7             | Video encoding method                                                  | BYTE      | See table 19                                    |
| 8             | Maximum number of audio physical channels supported by the terminal    | BYTE      |                                                 |
| 9             | Maximum number of physical channels of video supported by the terminal | BYTE      |                                                 |

Table 12 Definition of audio and video coding types

| Code | Name     | Remarks |
|------|----------|---------|
| 0    | reserved |         |

|   |        |       |
|---|--------|-------|
| 1 | G. 721 | audio |
| 2 | G. 722 | audio |
| 3 | G. 723 | audio |
| 4 | G. 728 | audio |
| 5 | G. 729 | audio |

Table 12 (continued)

| Code       | Name                             | Remarks  |
|------------|----------------------------------|----------|
| 6          | G.711A                           | audio    |
| 7          | G.711U                           | audio    |
| 8          | G.726                            | audio    |
| 9          | G.729A                           | audio    |
| 10         | DVI4_3                           | audio    |
| 11         | DVI4_4                           | audio    |
| 12         | DVI4_8K                          | audio    |
| 13         | DVI4_16K                         | audio    |
| 14         | LPC                              | audio    |
| 15         | S16BE_STEREO                     | audio    |
| 16         | S16BE_MONO                       | audio    |
| 17         | MPEGAUDIO                        | audio    |
| 18         | LPCM                             | audio    |
| 19         | AAC                              | audio    |
| 20         | WMA9STD                          | audio    |
| 21         | HEAAC                            | audio    |
| 22         | PCM_VOICE                        | audio    |
| 23         | PCM_AUDIO                        | audio    |
| 24         | AACLC                            | audio    |
| 25         | MP3                              | audio    |
| 26         | ADPCMA                           | audio    |
| 27         | MP4AUDIO                         | audio    |
| 28         | reserve                          | audio    |
| 29 to 90   | reserved                         |          |
| 91         | transmission (of<br>information) | systems  |
| 92 to 97   | reserved                         | video    |
| 98         | H.264                            | video    |
| 99         | H.265                            | video    |
| 100        | AVS                              | video    |
| 101        | SVAC                             | video    |
| 102 to 110 |                                  | reserved |

|            |  |              |
|------------|--|--------------|
| 111 to 127 |  | customizable |
|------------|--|--------------|

## 5.4 Video Alarm Commands

## 5.4.1 Video alarm reporting

The video alarm reporting adopts the method of reporting simultaneously with the position information, as the additional information of 0x0200 position information reporting, the definition table of additional information in Table 20 of JT/T 808-2011 is expanded, and the expanded definition of additional information is shown in Table 13.

Table 13 Additional Information Definition Table Extension

| Additional Information ID | Additional Information Length | Description and Requirements                                                                                                                                                                                                                     |
|---------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x14                      | 4                             | Video Related Alarms, DWORD, set by bit, see Table 14 for flag bit definitions                                                                                                                                                                   |
| 0x15                      | 4                             | Video Signal Loss Alarm Status, DWORD, set by bit, bit0 to bit31 indicate the 1st to 32nd logical channels respectively, if the corresponding bit is 1, it means that video signal loss occurs in this logical channel.                          |
| 0x16                      | 4                             | Video Signal Blocking Alarm Status, DWORD, set by bit, bit0 to bit31 indicate the 1st to 32nd logical channels respectively, if the corresponding bit is 1, it means that video signal blocking occurs in this logical channel.                  |
| 0x17                      | 2                             | Memory failure alarm status, WORD, set by bit, bit0 to bit11 indicate the 1st to 12th main memory, bit12 to bit15 indicate the 1st to 4th disaster recovery storage device, and the corresponding bit is 1, it means that the memory has failed. |
| 0x18                      | 2                             | Abnormal Driving Behavior Alarm Detailed Description, WORD, Definitions see Table 15                                                                                                                                                             |

Table 14 Video alarm flag bit definitions

| Bit  | definition                                          | Description of the process                                  |
|------|-----------------------------------------------------|-------------------------------------------------------------|
| 0    | Video signal loss alarm                             | Flag is maintained until the alarm condition is removed     |
| 1    | Video signal blocking alarm                         | Flag is maintained until the alarm condition is removed     |
| 2    | Memory Unit Failure Alarm                           | The flag is maintained until the alarm condition is removed |
| 3    | Other video equipment failure alarms                | Flag is maintained until the alarm condition is removed     |
| 4    | Bus overcrowding alarm                              | Flag is maintained until the alarm condition is removed     |
| 5    | Abnormal Driving Behavior Alarm                     | Flag is maintained until the alarm condition is removed     |
| 6    | Special alarm video reaches storage threshold alarm | Zeroed after receiving an answer                            |
| 7-31 | reserve                                             |                                                             |



Table 15 Definitions of Abnormal Driving Behavior Flag Positions

| Starting Byte | field                              | data type | Description and Requirements                                                                                                                |
|---------------|------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 0             | Types of Abnormal Driving Behavior | WORD      | Set by bit: 0 for no, 1 for yes;<br>bit0: Fatigue;<br>bit1: Call;<br>bit2: Smoking;<br>bit3 ~ bit10: Reserved;<br>bit11 ~ bit15: Customized |
| 2             | degree of fatigue                  | BYTE      | The degree of fatigue is expressed as 0 to 100, with larger values indicating more severe fatigue.                                          |

#### 5.4.2 Terminals uploading passenger traffic

Message ID: 0x1005.

Message type: signaling data message.

The terminal device counts the boarding and alighting passengers through video analysis and sends the counting result to the platform, and the data format of the message body is shown in Table 16.

Table 16 Terminal Upload Passenger Traffic Data Format

| Starting Byte | field                                  | data type | Description and Requirements                                                                       |
|---------------|----------------------------------------|-----------|----------------------------------------------------------------------------------------------------|
| 0             | starting time                          | BCD[6]    | YY-MM-DD-HH-MM-SS (GMT + 8 time, this time zone is used for all subsequent times in this standard) |
| 6             | end time                               | BCD[6]    | yy-mm-dd-hh-mm-ss                                                                                  |
| 12            | Number of people on board              | WORD      | Number of boardings from start time to end time                                                    |
| 14            | drop-off number (for a bus, train etc) | WORD      | Number of alightings from start time to end time                                                   |

#### 5.5 Real-time audio and video transmission commands

##### 5.5.1 Real-time audio and video transmission requests

Message ID: 0x9101.

Message type: signaling data message.

The platform requests real-time audio and video transmission from the terminal equipment, including real-time video transmission, initiating two-way voice intercom, one-way listening, broadcasting voice to all terminals and specific passthroughs. The message body data format is shown in Table 17. After receiving this message, the terminal replies to the general answer of the video terminal and then establishes a transmission link through the corresponding server IP address and port number, and then transmits the corresponding audio and video streaming data in accordance with the audio and video streaming transmission protocol.

Table 17 Real-time audio and video transmission request data formats

| Starting Byte | field                                            | data type | Description and Requirements      |
|---------------|--------------------------------------------------|-----------|-----------------------------------|
| 0             | Server IP address length                         | BYTE      | Length n                          |
| 1             | Server IP Address                                | STRING    | Live Video Server IP Address      |
| 1+n           | Server video channel listening port number (TCP) | WORD      | Live Video Server TCP Port Number |
| 3+n           | Server video channel listening port number (UDP) | WORD      | Live Video Server UDP Port Number |

|       |                        |      |                                                                                                                |
|-------|------------------------|------|----------------------------------------------------------------------------------------------------------------|
| 5 + n | logical channel number | BYTE | According to Table 2 in JT/T 1076-2016                                                                         |
| 6 + n | data type              | BYTE | 0: audio/video, 1: video, 2: two-way intercom, 3: listening, 4: center broadcast, 5: over-the-air transmission |
| 7 + n | stream type            | BYTE | 0: master stream, 1: sub-streams                                                                               |

When the platform receives a special alarm from the video terminal, it should take the initiative to issue this instruction without waiting for manual confirmation and start real-time audio and video transmission.

## 5.5.2 Audio and video real-time transmission control

Message ID: 0x9102.

Message type: signaling data message.

The platform sends audio/video real-time transmission control commands for switching code streams, suspending code stream transmission, closing audio/video transmission channels, etc. The message body data format is shown in Table 18.

Table 18 Audio and video real-time transmission control data formats

| Starting Byte | field                          | data type | Description and Requirements                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|--------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0             | logical channel number         | BYTE      | According to Table 2 in JT/T 1076-2016                                                                                                                                                                                                                                                                                                                                                              |
| 1             | control command                | BYTE      | <p>This command allows the platform to control the real-time audio and video of the device:</p> <p>0: Turns off the audio/video transmission command;</p> <p>1: Switching streams (adding pause and continue);</p> <p>2: Suspend all streams for this channel;</p> <p>3: Resume sending the pre-pause stream with the same stream type as before the pause;</p> <p>4: Turn off two-way intercom</p> |
| 2             | Turn off audio and video types | BYTE      | <p>0: Turns off the audio and video data related to this channel;</p> <p>1: Turns off only the audio associated with the channel and keeps the video associated with the channel;</p> <p>2: Turn off only the video related to the channel and keep the audio related to the channel.</p>                                                                                                           |
| 3             | Switching Stream Type          | BYTE      | <p>Switches the previously requested stream to the newly requested stream, with the audio remaining the same as before the switch.</p> <p>The new application has a code stream of:</p> <p>0: Main code stream;</p> <p>1: Subcode streams</p>                                                                                                                                                       |

## 5.5.3 Real-time audio and video streaming and passthrough data transmission

Message type: code stream data message.

The real-time audio/video streaming data transmission refers to the RTP protocol and is carried by UDP or TCP. The load packet format is based on the definition of IETF RFC 3550 RTP with additional fields such as message stream number, SIM card number, audio/video channel number, etc. The load packet format is defined in Table 19. The bit bits defined in the table are filled in according to the big-endian mode.

Table 19 Audio and video streaming and passthrough data transmission protocol load packet format definition table

| Starting Byte | field                 | data type | Description and Requirements |
|---------------|-----------------------|-----------|------------------------------|
| 0             | header identification | DWORD     | Fixed to 0x30 0x31 0x63 0x64 |
| 4             | V                     | 2 BITS    | Fixed to 2                   |
|               | P                     | 1 BIT     | Fixed to 0                   |

Table 19 (continued)

| Starting Byte | field                          | data type | Description and Requirements                                                                                                                                                                                              |
|---------------|--------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | X                              | 1 BIT     | Whether the RTP header requires expansion bits, fixed to 0                                                                                                                                                                |
|               | CC                             | 4 BITS    | Fixed at 1                                                                                                                                                                                                                |
| 5             | M                              | 1 BIT     | Flag bit that determines if it is the boundary of a complete data frame                                                                                                                                                   |
|               | PT                             | 7 BITS    | Load type, See Table 19                                                                                                                                                                                                   |
| 6             | package number                 | WORD      | Initially 0, for each RTP packet sent, the sequence number is increased by 1.                                                                                                                                             |
| 8             | SIM card number                | BCD[6]    | Terminal device SIM card number                                                                                                                                                                                           |
| 14            | logical channel number         | BYTE      | According to Table 2 in JT/T 1076-2016                                                                                                                                                                                    |
| 15            | data type                      | 4 BITS    | 0000: Video I frame;<br>0001: Video P-frame;<br>0010: Video B-frame;<br>0011: Audio Frame;<br>0100: Transmission of data                                                                                                  |
|               | Subcontract processing markers | 4 BITS    | 0000: Atomic packets, which may not be split;<br>0001: The first package in the subcontracting process;<br>0010: The last package in the subcontracting process;<br>0011: Intermediate packages in subcontract processing |
| 16            | timestamp                      | BYTE[8]   | Identifies the relative time in milliseconds (ms) of the current frame of this RTP packet. This field is not present when the data type is 0100.                                                                          |
| 24            | Last I Frame Interval          | WORD      | The time interval between this frame and the previous keyframe, in milliseconds (ms), or none when the data type is non-video frame                                                                                       |
| 26            | Last Frame Interval            | WORD      | The time interval between this frame and the previous frame in milliseconds (ms), which is not present when the data type is a non-video frame                                                                            |
| 28            | Data body length               | WORD      | Length of subsequent data body without this field                                                                                                                                                                         |
| 30            | data body                      | BYTE[ n ] | Audio/video data or transmission data up to 950 bytes in length                                                                                                                                                           |

#### 5.5.4 Real-time audio and video transmission status notification

Message ID: 0x9105.

Message type: signaling data message.

The platform sends a notification packet to the terminal according to the set time interval in the process of receiving audio and video data uploaded from the terminal, and the data format of the message body is shown in Table 20.

Table 20 Real-time audio and video transmission status notification data format

| Starting Byte | field                  | data type | Description and Requirements                                                                                         |
|---------------|------------------------|-----------|----------------------------------------------------------------------------------------------------------------------|
| 0             | logical channel number | BYTE      | According to Table 2 in JT/T 1076-2016                                                                               |
| 1             | packet loss            | BYTE      | The packet loss rate of the current transmission channel, multiplied by 100 and rounded to the nearest whole number. |

## 5.6 Historical audio and video query, playback and download commands

### 5.6.1 Querying the resource list

Message ID: 0x9205.

Message type: signaling data message.

The platform queries the list of video files from the terminal according to a combination of conditions such as audio/video type, channel number, alarm type and start/end time. The message body data format is shown in Table 21.

Table 21 Data format for querying the list of video files

| Starting Byte | field                              | data type | Description and Requirements                                                                                                                  |
|---------------|------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 0             | logical channel number             | BYTE      | According to Table 2 in JT/T 1076-2016, 0 indicates all channels.                                                                             |
| 1             | Starting time                      | BCD[6]    | YYY-MM-DD-HH-MM-SS, all 0 means no start time condition                                                                                       |
| 7             | end time                           | BCD[6]    | YY-MM-DD-HH-MM-SS, all 0 means no end time condition                                                                                          |
| 13            | warning sign                       | 64BITS    | bit0 ~ bit31 See JT/T 808-2011 Table 18 Alarm flag bit definition;<br>See Table 13 for bit32 to bit63;<br>All 0 means no alarm type condition |
| 21            | Types of audio and video resources | BYTE      | 0: audio and video, 1: audio, 2: video, 3: video or audio-video                                                                               |
| 22            | stream type                        | BYTE      | 0: all streams, 1: main stream, 2: sub-streams                                                                                                |
| 23            | Memory Type                        | BYTE      | 0: all memories, 1: main memory, 2: disaster recovery memory                                                                                  |

### 5.6.2 Terminal Upload Audio/Video Resource List

Message ID: 0x1205.

Message type: signaling data message.



The terminal responds to the platform's command to query the list of audio and video resources with the message that the terminal uploads the list of audio and video resources. If the list is too large and needs to be transmitted in packets, JT/T 808-2011 4.4.4.1.1 is used. 4.4.4.3 of JT/T 808-2011, the platform shall respond to each individual sub-packet with a common response of the video platform. The message body data format is shown in Table 22.

Table 22 Data format of the list of audio and video resources uploaded by terminals

| Starting Byte | field                           | data type | Description and Requirements                                                    |
|---------------|---------------------------------|-----------|---------------------------------------------------------------------------------|
| 0             | Serial Number                   | WORD      | Corresponds to the stream number of the query audio/video resource list command |
| 2             | Total audio and video resources | DWORD     | No eligible audio/video resources, set to 0                                     |
| 6             | Audio/Video Resource List       |           | See table 23                                                                    |

Table 23 List format of audio and video resources uploaded by terminals

| Starting Byte | field                              | data type | Description and Requirements                                                                |
|---------------|------------------------------------|-----------|---------------------------------------------------------------------------------------------|
| 0             | logical channel number             | BYTE      | According to Table 2 in JT/T 1076-2016                                                      |
| 1             | Starting time                      | BCD[6]    | yy-mm-dd-hh-mm-ss                                                                           |
| 7             | end time                           | BCD[6]    | yy-mm-dd-hh-mm-ss                                                                           |
| 13            | warning sign                       | 64BITS    | bit0 to bit31 are defined in Table 18 of JT/T 808-2011;<br>See Table 13 for bit32 to bit63. |
| 21            | Types of audio and video resources | BYTE      | 0: audio and video, 1: audio, 2: video                                                      |
| 22            | stream type                        | BYTE      | 1: main stream, 2: sub-streams                                                              |
| 23            | Memory Type                        | BYTE      | 1: main storage, 2: disaster storage recovery                                               |
| 24            | file size                          | DWORD     | Unit Byte (BYTE)                                                                            |

### 5.6.3 Remote video playback requests from the Platform

Message ID: 0x9201.

Message type: signaling data message.

When the platform requests audio and video recording playback from the terminal equipment, the terminal should respond with the command 0x1205 (terminal upload video file list), and then transmit the video data in the packet format defined in Table 18 Real-time Audio and Video Streaming Data Transmission RTP Protocol Load Packet Format. The message body data format is shown in Table 24.

Table 24 Data format for remote video playback requests from the platform

| Starting Byte | field                    | data type | Description and Requirements            |
|---------------|--------------------------|-----------|-----------------------------------------|
| 0             | Server IP address length | BYTE      | Length n                                |
| 1             | Server IP Address        | STRING    | Real-Time Audio/Video Server IP Address |

|     |                                                        |      |                                                                                       |
|-----|--------------------------------------------------------|------|---------------------------------------------------------------------------------------|
| 1+n | Server audio/video channel listening port number (TCP) | WORD | Real-time audio/video server port number, set to 0 when not using TCP transmission.   |
| 3+n | Server audio/video channel listening port number (UDP) | WORD | Real-time audio/video server port number, set to 0 when UDP transmission is not used. |
| 5+n | logical channel number                                 | BYTE | According to Table 2 in JT/T 1076-2016                                                |

Table 24 (continued)

| Starting Byte | field                                   | data type | Description and Requirements                                                                                                                   |
|---------------|-----------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 + n         | Audio/Video Type                        | BYTE      | 0: audio and video, 1: audio, 2: video, 3: video or audio-video                                                                                |
| 7 + n         | stream type                             | BYTE      | 0: main stream or sub-stream, 1: main stream, 2: sub-stream; if this channel only transmits audio, this field is set to 0                      |
| 8 + n         | Memory Type                             | BYTE      | 0: main memory or disaster recovery memory, 1: main memory, 2: disaster recovery memory                                                        |
| 9 + n         | playback method                         | BYTE      | 0: Normal playback;<br>1: Fast-forward through the replay;<br>2: Keyframe snapback release;<br>3: Keyframe playback;<br>4: Single frame upload |
| 10 + n        | Fast Forward or Fast Reverse Multiplier | BYTE      | The contents of this field are valid when the playback mode is 1 and 2, otherwise it is set to 0.<br>0: Invalid;<br>1: 1 times;<br>2: 2 times  |
| 10 + n        | Fast Forward or Fast Reverse Multiplier | BYTE      | 3: 4 times;<br>4: 8 times;<br>5: 16 times                                                                                                      |
| 11 + n        | Starting time                           | BCD[6]    | YYY-MM-DD-HH-MM-SS, when playback mode is 4, this field indicates the single frame upload time                                                 |
| 17 + n        | end time                                | BCD[6]    | YYY-MM-DD-HH-MM-SS, 0 means playback all the time, this field is invalid when playback mode is 4                                               |

#### 5.6.4 Remote video playback control from the platform

Message ID: 0x9202.

Message type: signaling data message.

During the playback of audio and video recordings by the terminal equipment, the platform can issue playback control commands to control the playback process. The message body data format is shown in Table 25.

Table 25 Format for remote video playback control data sent from the platform

| Starting Byte | field                      | data type | Description and Requirements           |
|---------------|----------------------------|-----------|----------------------------------------|
| 0             | Audio/Video Channel Number | BYTE      | According to Table 2 in JT/T 1076-2016 |

|   |                  |      |                                                                                                                                                                                                |
|---|------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Playback control | BYTE | 0: Start playback;<br>1: Pause for playback;<br>2: End playback;<br>3: Fast-forward through the replay;<br>4: Keyframe snapback release;<br>5: Drag and drop playback;<br>6: Keyframe playback |
|---|------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table 25 (continued)

| Starting Byte | field                                   | data type | Description and Requirements                                                                                                                                                              |
|---------------|-----------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2             | Fast Forward or Fast Reverse Multiplier | BYTE      | The contents of this field are valid when the playback control is 3 and 4, otherwise it is set to 0.<br>0: Invalid;<br>1:1 times;<br>2:2 times;<br>3:4 times;<br>4:8 times;<br>5:16 times |
| 3             | Drag the playback position              | BCD[6]    | YY-MM-DD-HH-MM-SS, valid if playback control is 5                                                                                                                                         |

## 5.6.5 File Upload Instructions

Message ID: 0x9206.

Message type: signaling data message.

The platform sends a file upload command to the terminal, and the terminal replies with a generic response and uploads the file to the specified path of the target FTP server via FTP. The message body data format is shown in Table 26.

Table 26 File Upload Command Data Format

| Starting Byte | field                   | data type | Description and Requirements                                                                                                                 |
|---------------|-------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 0             | Server address length   | BYTE      | Length k                                                                                                                                     |
| 1             | server address          | STRING    | FTP Server Address                                                                                                                           |
| 1+k           | ports                   | WORD      | FTP Server Side Slogan                                                                                                                       |
| 3+k           | User name length        | BYTE      | Length l                                                                                                                                     |
| 4+k           | user ID                 | STRING    | FTP User Name                                                                                                                                |
| 4+k+l         | Password length         | BYTE      | Length m                                                                                                                                     |
| 5+k+l         | cryptographic           | STRING    | FTP Password                                                                                                                                 |
| 5+k+l+m       | File upload path length | BYTE      | Length n                                                                                                                                     |
| 6+k+l+m       | File Upload Path        | STRING    | File Upload Path                                                                                                                             |
| 6+k+l+m+n     | logical channel number  | BYTE      | See Table 2 in JT/T 1076-2016                                                                                                                |
| 7+k+l+m+n     | Starting time           | BCD[6]    | yy-mm-dd-hh-mm-ss                                                                                                                            |
| 13+k+l+m+n    | end time                | BCD[6]    | yy-mm-dd-hh-mm-ss                                                                                                                            |
| 19+k+l+m+n    | warning sign            | 64BITS    | bit0 ~ bit31 See JT/T 808-2011 Table 18 Alarm flag bit definition;<br>See Table 12 for bit32 to bit63;<br>All 0 means do not specify whether |

|              |                                    |      |                                                             |
|--------------|------------------------------------|------|-------------------------------------------------------------|
|              |                                    |      | there is an alarm or not                                    |
| $27+k+l+m+n$ | Types of audio and video resources | BYTE | 0:audio and video, 1:audio, 2:video, 3:video or audio-video |

Table 26 (continued)

| Starting Byte | field                     | data type | Description and Requirements                                                                                                                                                                                                                                         |
|---------------|---------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28+k+l+m+n    | stream type               | BYTE      | 0: main stream or sub-stream, 1: main stream, 2: sub-streams                                                                                                                                                                                                         |
| 29+k+l+m+n    | storage location          | BYTE      | 0: main memory or disaster recovery memory, 1: main memory, 2: disaster recovery memory                                                                                                                                                                              |
| 30+k+l+m+n    | Task execution conditions | BYTE      | Expressed in bit bits:<br>bit0: WIFI, when it is 1, it means download is available under WI-FI;<br>bit1: LAN, when it is 1, it means download is available when LAN is connected;<br>bit2: 3G/4G, when it is 1, it means download is available for 3G/4G connection. |

## 5.6.6 Notification of completion of document upload

Message ID: 0x1206.

Message type: signaling data message.

When all the files are uploaded via FTP, the terminal reports this command to notify the platform. The message body data format is shown in Table 27.

Table 27 File upload completion notification data format

| Starting Byte | field                       | data type | Description and Requirements                                           |
|---------------|-----------------------------|-----------|------------------------------------------------------------------------|
| 0             | response number (computing) | WORD      | Corresponds to the stream number of the platform's file upload message |
| 2             | in the end                  | BYTE      | 0: Success;<br>1: Failure                                              |

## 5.6.7 File upload control

Message ID: 0x9207.

Message type: signaling data message.

The platform notifies the terminal to pause, continue, or cancel all files in transit . The message body data format is shown in Table 28.

Table 28 File Upload Control Data Format

| Starting Byte | field                       | data type | Description and Requirements                                   |
|---------------|-----------------------------|-----------|----------------------------------------------------------------|
| 0             | response number (computing) | WORD      | Corresponds to the stream number of the platform's file upload |



|   |                |      |                                                   |
|---|----------------|------|---------------------------------------------------|
|   |                |      | message                                           |
| 2 | upload control | BYTE | 0: Suspension;<br>1: Continue;<br>2: Cancellation |

## 5.7 Head Control Commands

### 5.7.1 Head Rotation

Message ID: 0x9301.

Message type: signaling data message.

The platform requests the rotating lens from the terminal . The message body data format is shown in Table 29.

Table 29 Head Rotation Data Format

| Starting Byte | field                  | data type | Description and Requirements                           |
|---------------|------------------------|-----------|--------------------------------------------------------|
| 0             | logical channel number | BYTE      | According to Table 2 in JT/T 1076-2016                 |
| 1             | orientations           | BYTE      | 0: Stop;<br>1: Up;<br>2: Down;<br>3: Left;<br>4: Right |
| 2             | tempo                  | BYTE      | 0 to 255                                               |

#### 5.7.2 Head Adjustment Focus Control

Message ID: 0x9302.

Message type: signaling data message.

The platform requests the terminal to adjust the lens focus . The message body data format is shown in Table 30.

Table 30 Head Adjustment Lens Focus Control Data Format

| Starting Byte | field                      | data type | Description and Requirements                                 |
|---------------|----------------------------|-----------|--------------------------------------------------------------|
| 0             | logical channel number     | BYTE      | According to Table 2 in JT/T 1076-2016                       |
| 1             | Focus Adjustment Direction | BYTE      | 0: Increase the focal length<br>1: Decrease the focal length |

#### 5.7.3 Head Adjustment Aperture Control

Message ID: 0x9303.

Message type: signaling data message.

The platform requests the terminal to adjust the lens aperture . The message body data format is shown in Table 31.

Table 31 Head Adjustment Lens Aperture Control Data Format

| Starting Byte | field                      | data type | Description and Requirements           |
|---------------|----------------------------|-----------|----------------------------------------|
| 0             | logical channel number     | BYTE      | According to Table 2 in JT/T 1076-2016 |
| 1             | Aperture adjustment method | BYTE      | 0: Turn it up<br>1: Turn it down       |

#### 5.7.4 Head Wiper Control

Message ID: 0x9304.

Message type: signaling data message.

The platform requests wipers from the terminal . The message body data format is shown in Table 32.

Table 32 Head Wiper Control Data Format

| Starting Byte | field                  | data type | Description and Requirements           |
|---------------|------------------------|-----------|----------------------------------------|
| 0             | logical channel number | BYTE      | According to Table 2 in JT/T 1076-2016 |
| 1             | start-stop marking     | BYTE      | 0:Stop;<br>1:Initiation                |

## 5.7.5 Infrared fill light control

Message ID: 0x9305.

Message type: signaling data message.

The platform requests infrared fill light control from the terminal . The message body data format is shown in Table 33.

Table 33 Infrared fill light control data format

| Starting Byte | field                  | data type | Description and Requirements           |
|---------------|------------------------|-----------|----------------------------------------|
| 0             | logical channel number | BYTE      | According to Table 2 in JT/T 1076-2016 |
| 1             | start-stop marking     | BYTE      | 0:Stop;<br>1:Initiation                |

## 5.7.6 Head Variable Control

Message ID: 0x9306.

Message type: signaling data message.

The platform requests multiplication control from the terminal . The message body data format is shown in Table 34.

Table 34 Head Variable Control Data Format

| Starting Byte | field                  | data type | Description and Requirements           |
|---------------|------------------------|-----------|----------------------------------------|
| 0             | logical channel number | BYTE      | According to Table 2 in JT/T 1076-2016 |
| 1             | variable speed control | BYTE      | 0:turn up; 1:turn down                 |

## 5.8 Terminal Sleep Wakeup Command

The platform sends wake-up SMS to wake up the dormant terminal to start working, the content of the SMS is "WAKEUPXX", where XX represents the length of wake-up, the unit is minute, and the value range is 0 to 65,536. If the value is 0, it means that the terminal is always in wake-up state until the terminal ACC ON or lower than the rated voltage.

## 6 Streaming communication between audio and video streaming servers and client playback software

## 6.1 Audio-video streaming and pass-through data encapsulation formats

See Table 18 for definitions of audio and video streams and passthrough data encapsulation formats between the video platform and the client playback software.

#### 6.2 Audio and Video Streaming Request URL Command Format

The government video monitoring platform sends a real-time preview or remote playback request command to the enterprise video monitoring platform and gets a successful answer, then it gets the

Get the IP address and port number of the audio/video streaming server, and the client of the government video monitoring platform sends a URL command directly to the enterprise audio/video streaming server, establishes a link to get the audio/video streaming data, and the client plays it through a browser plug-in or special software.

The audio/video stream request URL should not be displayed in the interface. The command format is defined as follows:

http:// [Server IP Address] : [Port Number]/ [License Plate Number] . [License Plate Color] . [Logical Channel Number] . [Audio/Video Flag] . [timeout password] .

The data items of the Audio/Video Streaming Request URL command are defined in Table 35.

Table 35 Definition table for data items of audio/video stream request URL command

| field                         |                           | Description and Requirements                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Address Attribute Information | Server IP Address         | Audio/Video Streaming Server IP Address                                                                                                                                                                                                                                                                                                                                                         |
|                               | port number               | Audio/Video Streaming Service Port Number                                                                                                                                                                                                                                                                                                                                                       |
|                               | license plate number      | Should be UTF-8 encoded and converted to the application/x-www-form-urlencoded MIME format of IETF RFC 2854                                                                                                                                                                                                                                                                                     |
|                               | license plate color       | 5. 4. 12 in accordance with JT/T 415-2006. 4. 12 of JT/T 415-2006                                                                                                                                                                                                                                                                                                                               |
|                               | logical channel number    | According to Table 2 in JT/T 1076-2016, 0 indicates all channels.                                                                                                                                                                                                                                                                                                                               |
|                               | sound and video signage   | 0: Audio and video;<br>1: Audio;<br>2: Video                                                                                                                                                                                                                                                                                                                                                    |
| Additional Information        | password (used by sentry) | Generated by the server of the enterprise platform, the time password of the client of the government platform of the attributed region is different from that of the government platform of the inter-regional region. The time password shall be composed of English letters (including case) and Arabic numerals only, with a length of 64 ASCII characters, and shall be updated every 24h. |
|                               | location marking          | The satellite positioning time and latitude/longitude of the vehicle at any time within 5 minutes, used for verification when accessing the government platform in the cross-domain area, and can be empty when accessing the client of the government platform in the attributed area. expressed in ASCII characters, in the format of:<br>yyymmdd - hhmmss - nxx. xxxxxx - exxx. xxxxxx       |

## 7 Basics of communication protocols between video platforms

The communication methods, data types, security authentication methods and protocol message formats between different video platforms are in accordance with the requirements of Chapter 4 of JT/T 809-2011.

Data transmission between different video platforms does not require authentication, and the transmission channel should use the link that has already been established between the positioning platforms, without adding new links.

## 8 Communication protocol flow between video platforms

### 8.1 Operational class for timeout password reporting and requesting

The time limit password is automatically generated by the enterprise video monitoring platform every day and uploaded to the video monitoring platform of the government of the place of belonging on its own initiative, and the video monitoring platform of the government of the place of belonging uploads it to the video monitoring platform of the government of the place of belonging on its own initiative after it receives the time limit password of the day. When the cross-domain government video monitoring platform needs to access the audio and video information of cross-domain vehicles, it should request the cross-domain time-expired password of the day from the video monitoring platform of the higher-level government, and the video monitoring platform of the higher-level government will give a corresponding response according to the geographical location of the vehicles after receiving the request.

## 8.2 Real-time audio and video services

### 8.2.1 Real-time uploading of audio and video data from enterprise video surveillance platforms to government video surveillance platforms

The government video surveillance platform sends real-time audio and video upload requests to the enterprise video surveillance platform, and the enterprise video surveillance platform receives the requests and should give an answer to the government video surveillance platform. If the answer is successful, the government video monitoring platform requests real-time audio and video data from the IP and port of the video server specified by the enterprise video monitoring platform.

### 8.2.2 The enterprise video surveillance platform stopped uploading audio and video data in real time to the government video surveillance platform

The government video surveillance platform sends a request to the enterprise video surveillance platform to stop uploading real-time audio and video, and the enterprise video surveillance platform receives the request and should give an answer to the government video surveillance platform. If the answer is successful, the enterprise video surveillance platform stops sending real-time audio and video data to the government video surveillance platform.

## 8.3 Remote video retrieval business class

### 8.3.1 Directory of audio and video resources obtained by government video surveillance platforms from enterprise video monitoring platforms

The government video monitoring platform sends a request to the enterprise video monitoring platform to obtain the audio/video resource catalog. After receiving the request, the enterprise video monitoring platform should immediately retrieve the latest audio/video resource catalog from the terminals and update the local catalog to answer to the government video monitoring platform. If the answer is successful, the enterprise video monitoring platform sends the audio and video resource catalog data to the government video monitoring platform.

### 8.3.2 Directory of audio and video resources proactively uploaded by enterprise video surveillance platforms to government video surveillance platforms

After the enterprise video monitoring platform receives the special alarm information uploaded by the terminal and waits for the complete recording of the video information, it shall retrieve the latest audio and video resource catalogs with special alarm identifiers from the terminal and update the local catalogs and then upload the audio and video resource catalogs to the governmental video monitoring platform on its own initiative.

## 8.4 Remote Video Download Service Class

### 8.4.1 Downloading video data from government video surveillance platforms to enterprise video surveillance platforms



The government video surveillance platform sends a request for video data to the enterprise video surveillance platform, and the enterprise video surveillance platform receives the request and should give an answer to the government video surveillance platform. If the answer is successful, the government video surveillance platform can request video data from the FTP server IP and port specified by the enterprise video surveillance platform.

#### 8.4.2 Enterprise video surveillance platforms send download completion notifications to government video surveillance platforms

The enterprise video surveillance platform sends a download completion notification to the government video surveillance platform, and when the government video surveillance platform receives the notification, it indicates that the video data has been downloaded from the terminal, and the government video surveillance platform can request video data from the IP and port of the video FTP server specified by the enterprise video surveillance platform.

#### 8.4.3 Government video monitoring platform sends download control commands to enterprise video monitoring platforms

The government video monitoring platform sends download control instructions to the enterprise video monitoring platform, and after receiving the instructions, the enterprise video monitoring platform should respond to the corresponding control actions in time and give an answer to the government video monitoring platform.

### 8.5 Remote video playback business class

#### 8.5.1 Government Video Surveillance Platforms Requesting Video Playback from Enterprise Video Surveillance Platforms

The government video monitoring platform sends a request for video playback to the enterprise video monitoring platform, and after receiving the request, the enterprise video monitoring platform should

Answer to the government video monitoring platform. If the answer result is successful, the government video monitoring platform requests historical audio and video streaming data from the IP and port of the audio and video streaming server specified by the enterprise video monitoring platform.

8.5.2 Government video surveillance platforms stop requesting video playback from enterprise video surveillance platforms

The government video surveillance platform sends a request to the enterprise video surveillance platform to stop playing back the video, and upon receiving the request, the enterprise video surveillance platform should give an answer to the government video surveillance platform and stop sending historical audio and video streaming data to the government video surveillance platform.

## 9 Video inter-platform communication protocol constant definitions

### 9.1 Operational data type identification

The names and identifiers of the service data types specified in the Audio/Video Data Exchange protocol are shown in Table 36.

Table 36 Cross-reference of operational data type names and identifiers

| Type of message                | Business Data Type Name                                          | message link | Business data type identification | numerical value |
|--------------------------------|------------------------------------------------------------------|--------------|-----------------------------------|-----------------|
| Timing password business class | Master Link Time Command Interaction Message                     | primary link | UP AUTHORIZE MSG                  | 0x1700          |
|                                | From Link Time Command Interaction Message                       | slave link   | DOWN AUTHORIZE MSG                | 0x9700          |
| Real-time Audio/Video Services | Real-time audio and video interactions on the main link Messages | primary link | UP REALVIDEO MSG                  | 0x1800          |
|                                | Real-time audio and video interaction from the link Message      | slave link   | DOWN REALVIDEO MSG                | 0x9800          |
| Remote video retrieval         | Main Link Remote Video Retrieval Interactive Messages            | primary link | UP SEARCH MSG                     | 0x1900          |
|                                | Retrieving Interactive Messages from Link Remote Recording       | slave link   | DOWN SEARCH MSG                   | 0x9900          |
|                                | Main Link Remote Video Playback                                  | primary link | UP PLAYBACK MSG                   | 0x1A00          |

|                       |                                                             |              |                   |        |
|-----------------------|-------------------------------------------------------------|--------------|-------------------|--------|
| Remote video playback | Interactive Messages                                        |              |                   |        |
|                       | Remote video playback from the link Interactive messages    | slave link   | DOWN PLAYBACK MSG | 0x9A00 |
| Remote Video Download | Main Link Remote Video Download Interactive Messages        | primary link | UP DOWNLOAD MSG   | 0x1B00 |
|                       | Downloading Interactive Messages from Link Remote Recording | slave link   | DOWN DOWNLOAD MSG | 0x9B00 |

## 9.2 Sub-service type identification

The names and identifiers of the sub-service types specified in the data exchange protocol are shown in Table 37.

Table 37 Sub-service Type Name and Identifier Comparison Table

| Business data types                                                            | Sub-business type name                                                        | Sub-service data type identification | value  |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------|--------|
| Main Link Time Password Service Class Message UP AUTHORIZE MSG                 | Timing passwords to report messages                                           | UP AUTHORIZE MSG STARTUP             | 0x1701 |
|                                                                                | Time Password Request Message                                                 | UP AUTHORIZE MSG STARTUP REQ         | 0x1702 |
| From Link Time Password Business Class Message DOWN BASE DATA MSG              | Time Password Request Answer Message                                          | DOWN AUTHORIZE MSG STARTUP REQ ACK   | 0x9702 |
| Main link real-time audio and video interactions Consumer UP REALVIDEO MSG     | Real-time audio and video request answer messages                             | UP REALVIDEO MSG STARTUP ACK         | 0x1801 |
|                                                                                | Unsolicited request to stop real-time audio/video transmission Answer message | UP REALVIDEO MSG END ACK             | 0x1802 |
| Real-time audio and video interaction from the link Message DOWN REALVIDEO MSG | Real-time audio and video requests Message                                    | DOWN REALVIDEO MSG STARTUP           | 0x9801 |
|                                                                                | Proactive request to stop real-time audio and video transmission messages     | DOWN REALVIDEO MSG END               | 0x9802 |
| Main Link Remote Video Retrieval Interaction Message UP SEARCH MSG             | Proactive upload of audio and video resource catalog information messages     | UP FILELIST MSG                      | 0x1901 |
|                                                                                | Query Audio/Video Resource Directory Answer Message                           | UP REALVIDEO FILELIST REQ ACK        | 0x1902 |
| Retrieving Interactive Messages from Link Remote Recording DOWN SEARCH MSG     | Proactive Upload Audio/Video Resource Catalog Information Answer Message      | DOWN FILELIST MSG ACK                | 0x9901 |

|                                                                              |                                                      |                                 |        |
|------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------|--------|
|                                                                              | Query Audio/Video Resource Catalog Request Message   | DOWN REALVIDEO FILELIST REQ     | 0x9902 |
| Main Link Remote Video Playback Interactive Messages UP PLAYBACK MSG         | Remote Video Playback Request Response Message       | up playback msg startup ack     | 0x1A01 |
|                                                                              | Remote video playback control Answer message         | up playback msg control ack     | 0x1A02 |
| Remote video playback from the link Interactive messages DOWN PLAY- BACK MSG | Remote Video Playback Request Message                | DOWN PLAYBACK MSG STARTUP       | 0x9A01 |
|                                                                              | Remote Video Playback Control Message                | DOWN PLAYBACK MSG CONTROL       | 0x9A02 |
| Main Link Remote Video Download Interactive Messages UP DOWNLOAD MSG         | Remote Video Download Request Response Message       | up download msg startup ack     | 0x1B01 |
|                                                                              | Remote video download complete Notification message  | UP DOWNLOAD MSG END INFORM      | 0x1B02 |
|                                                                              | Remote Video Download Control Answer Message         | UP DOWNLOAD MSG CONTROL ACK     | 0x1B03 |
| Download Interactive Messages from Link Remote Video DOWN- LOAD MSG          | Remote Video Download Request Message                | DOWN DOWNLOAD MSG STARTUP       | 0x9B01 |
|                                                                              | Remote video download complete Notify answer message | up down load msg end inform ack | 0x9B02 |
|                                                                              | Remote Video Download Control Message                | DOWN DOWNLOAD MSG CONTROL       | 0x9B03 |

### 9.3 Video alarm type coding

The coding of the types of video alarms reported through the platform is shown in Table 38.

Table 38 Vehicle Video Alarm Type Coding Table

| Code   | Name                                                | Description and Requirements |
|--------|-----------------------------------------------------|------------------------------|
| 0x0101 | Video signal loss alarm                             | —                            |
| 0x0102 | Video signal blocking alarm                         | —                            |
| 0x0103 | Memory Unit Failure Alarm                           | —                            |
| 0x0104 | Other video equipment failure alarms                | —                            |
| 0x0105 | Bus overcrowding alarm                              | —                            |
| 0x0106 | Abnormal Driving Behavior Alarm                     | —                            |
| 0x0107 | Special alarm video reaches storage threshold alarm | —                            |

## 10 Video inter-platform communication protocol data format

### 10.1 Time-limited password reporting and request business class

#### 10.1.1 Time-limited password reporting message

Link type: Main link

Message direction: From subordinate platform to superior platform

Sub-service type identifier: UP AUTHORIZE MSG STARTUP.

Description: The corporate video surveillance platform to the government video surveillance platform, or the subordinate government video surveillance platform to the superior government video surveillance platform, actively reports the time-limited password. See Table 39 for the data body. This command does not require a response.

Table 39 Time-limited password reporting message data body

| Field Name       | Number of bytes | Data Type | Description and Requirements                                                                                                                                                    |
|------------------|-----------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DATA TYPE        | 2               | uint16t   | Sub-business type identification                                                                                                                                                |
| PLATFORM ID      | 11              | BYTES     | Unique code for enterprise video surveillance platform, code of the administrative division of the enterprise to which the platform belongs + the platform announcement number. |
| AUTHORIZE CODE 1 | 64              | BYTES     | Time-sensitive passwords used by government platforms in the attributed regions                                                                                                 |

|                  |    |       |                                                                             |
|------------------|----|-------|-----------------------------------------------------------------------------|
| AUTHORIZE CODE 2 | 64 | BYTES | Time-sensitive passwords used by government platforms in cross-domain areas |
|------------------|----|-------|-----------------------------------------------------------------------------|

#### 10.1 . 2 Statute of limitations password request message

Link type: primary link

Message direction: cross-domain local government video monitoring platform to higher level government video monitoring platform

Sub-service type identifier: UP AUTHORIZE MSG STARTUP REQ.

Description: The cross-domain local government video surveillance platform obtains from the higher-level government video surveillance platform the current day's time-sensitive password of the enterprise's video surveillance platform where the designated vehicle is located, and the data body is shown in Table 40.

Table 40 Statute of limitations password request message dataset

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                 |
|---------------|-----------------|-------------|--------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                         |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006 |
| DATA TYPE     | 2               | uint16 t    | Sub-business type identification                             |
| DATA LENGTH   | 4               | uint32 t    | Length of subsequent data, value 0x00000000                  |

## 10.1 . 3 Statute of limitations password request reply message

Link type: slave link

Message direction: higher-level government video monitoring platform to cross-domain local government video monitoring platform

Sub-service type identification: DOWN AUTHORIZE MSG STARTUP REQ ACK .

Description: The higher level government video surveillance platform answers the time password request message sent by the cross-domain local government video surveillance platform, and the higher level government video surveillance platform determines the content of the answer according to the geographic location of the requesting vehicle within 5 minutes. The data body is shown in Table 41.

Table 41 Statute of limitations password request response message data body

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                 |
|---------------|-----------------|-------------|--------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                         |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006 |
| DATA TYPE     | 2               | uint16 t    | Sub-business type identification                             |
| DATA LENGTH   | 4               | uint32 t    | Length of data for the next 2 fields                         |

## 10.2 Real-time audio and video service classes

## 10.2 . 1 Real-time audio and video request messages

Link type: slave link

Message direction: originator platform to receiver platform

Sub-service type identification: DOWN REALVIDEO MSG STARTUP.

Description: The government video monitoring platform sends this command to the enterprise video monitoring platform, the higher-level government platform to the lower-level government platform, or the cross-domain regional government platform to the government platform of the belonging region to request the real-time audio and video of the vehicle . The data body is shown in Table 42.

Table 42 Real-time audio and video request datasets



| Field Name    | Number of bytes | Data Type   | Description and Requirements                                 |
|---------------|-----------------|-------------|--------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                         |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006 |
| DATA TYPE     | 2               | uint16 t    | Sub-business type identification                             |

Table 42 (continued)

| Field Name     | Number of bytes | Data Type | Description and Requirements                                                                                                                                                                                                       |
|----------------|-----------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DATA LENGTH    | 4               | uint32 t  | Length of data in subsequent 4 fields                                                                                                                                                                                              |
| CHANNEL ID     | 1               | BYTE      | Logical channel number, according to table 2 in JT/T 1076-2016, 0 means all channels                                                                                                                                               |
| AVITEM TYPE    | 1               | BYTE      | Audio and video types, defined as follows:<br>0x00: audio and video; 0x01: audio; 0x02: video                                                                                                                                      |
| AUTHORIZE CODE | 64              | BYTES     | password (used by sentry)                                                                                                                                                                                                          |
| GNSS DATA      | 36              | BYTES     | This field is only used for cross-domain access requests if the vehicle is located within 5 minutes of entering the cross-domain area, in accordance with JT/T 809-2011 protocol 4. 5. 8. 1. 5. 8. 1 of the JT/T 809-2011 protocol |

### 10.2.2 Real-time audio and video request reply messages

Link type: primary link

Message direction: receiver platform to originator platform

Sub-service type identification: UP REALVIDEO MSG STARTUP ACK.

Description: The enterprise video surveillance platform responds to the real-time audio and video request messages of vehicles sent by the government video monitoring platform. The data body is shown in Table 43.

Table 43 Real-time audio/video response datasets

| Field Name    | Number of bytes | Data Type    | Description and Requirements                                                                                                                                                                           |
|---------------|-----------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VEHICLE NO    | 21              | Octet String | license plate number                                                                                                                                                                                   |
| VEHICLE COLOR | 1               | BYTE         | Color of license plate, according to 5.4.12 of JT/T 415-2006                                                                                                                                           |
| DATA TYPE     | 2               | uint16 t     | Sub-business type identification                                                                                                                                                                       |
| DATA LENGTH   | 4               | uint32 t     | Length of data for the next 3 fields                                                                                                                                                                   |
| RESULT        | 1               | BYTE         | Answer results, defined as follows:<br>0x00: Success;<br>0x01: Failure;<br>0x02: Not supported;<br>0x03: End of session;<br>0x04: Timing password error;<br>0x05: Cross-domain condition not satisfied |

|             |    |             |                                     |
|-------------|----|-------------|-------------------------------------|
| SERVER IP   | 32 | OctetString | Enterprise Video Server IP Address  |
| SERVER PORT | 2  | uint16t     | Enterprise Video Server Side Slogan |

### 10.2.3 Unsolicited request to stop live audio/video transmission message

Link type: slave link

Message Direction: Government Video Supervision Platform to Enterprise Video Surveillance Platform

Sub-service type identification: DOWN-REALVIDEO-MSG-END.

DESCRIPTION: The government video monitoring platform sends this command to the enterprise video monitoring platform to initiate a request to stop the real-time audio and video transmission of the vehicle. The data body is shown in Table 44.

Table 44 Actively request to stop the real-time audio and video message data body.

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                                                  |
|---------------|-----------------|-------------|-----------------------------------------------------------------------------------------------|
| VEHICLE-NO    | 21              | OctetString | license plate number                                                                          |
| VEHICLE-COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006                                  |
| DATA-TYPE     | 2               | uint16-t    | Sub-business type identification                                                              |
| DATA-LENGTH   | 4               | uint32-t    | Length of data for the next 2 fields                                                          |
| CHANNEL-ID    | 1               | BYTE        | Logical channel number, according to table 2 in JT/T 1076-2016, 0 means all channels          |
| AVITEM-TYPE   | 1               | BYTE        | Audio and video types, defined as follows:<br>0x00: audio and video; 0x01: audio; 0x02: video |

10. 2 . 4 Actively request to stop the real-time audio and video transmission response message

11. Link type: primary link

Message direction: enterprise video surveillance platform to the government video monitoring platform

Sub-service type identifier: UP-REALVIDEO-MSG-END-ACK.

Description: The enterprise video surveillance platform responds to an unsolicited request sent by the government video surveillance platform to stop real-time audio and video transmission messages, see Table 45 for the data body.

Table 45 Actively request to stop the real-time audio and video response message data body

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                 |
|---------------|-----------------|-------------|--------------------------------------------------------------|
| VEHICLE-NO    | 21              | OctetString | license plate number                                         |
| VEHICLE-COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006 |
| DATA-TYPE     | 2               | uint16-t    | Sub-business type identification                             |
| DATA-LENGTH   | 4               | uint32-t    | Subsequent 1 field data length                               |

|        |   |      |                                                                                                                         |
|--------|---|------|-------------------------------------------------------------------------------------------------------------------------|
| RESULT | 1 | BYTE | Answer results, defined as follows:<br>0x00: Success;<br>0x01: Failure;<br>0x02: Not supported;<br>0x03: End of session |
|--------|---|------|-------------------------------------------------------------------------------------------------------------------------|

## 10.3 Remote video retrieval

## 10.3.1 Proactive Upload Audio/Video Resource Catalog Information Message

Link type: primary link

Message direction: lower platform to higher platform

Sub-service type identification: UP FILELIST MSG.

Description: The enterprise video monitoring platform sends the audio and video resource catalog with special alarm mark to the government video monitoring platform, or the lower level government platform sends the higher level government platform on its own initiative. The data body is shown in Table 46.

Table 46 Active Upload Audio and Video Resource Catalog Data Bodies

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                 |
|---------------|-----------------|-------------|--------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                         |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006 |
| DATA TYPE     | 2               | uint16 t    | Sub-business type identification                             |
| DATA LENGTH   | 4               | uint32 t    | Length of data for the next 2 fields                         |
| ITEM NUM      | 4               | uint32 t    | Total number of resource catalog entries                     |
| ITEM LIST     |                 |             | For a list of resource catalog entries, see Table 47         |

Table 47 Format of uploaded audio/video resource catalog item listings

| Field Name  | Number of bytes | Data Type | Description and Requirements                                                                   |
|-------------|-----------------|-----------|------------------------------------------------------------------------------------------------|
| CHANNEL ID  | 1               | BYTE      | Logical channel number, according to table 2 in JT/T 1076-2016, 0 means all channels           |
| START TIME  | 8               | time t    | UTC time                                                                                       |
| END TIME    | 8               | time t    | UTC time                                                                                       |
| ALARM TYPE  | 8               | 64BITS    | bit0-31 are defined according to JT/T 808-2011 Table 18 Alarm flag bits; bit32-63 See table 10 |
| AVITEM TYPE | 1               | BYTE      | Audio and video types, defined as follows:<br>0x00: audio and video; 0x01: audio; 0x02: video  |
| STREAM TYPE | 1               | BYTE      | The code stream type, defined as follows:<br>0x01: main stream; 0x02: sub-streams              |

|           |   |          |                                                                              |
|-----------|---|----------|------------------------------------------------------------------------------|
| MEM TYPE  | 1 | BYTE     | memory type, defined as follows:<br>0x01: main memory; 0x02: disaster memory |
| FILE SIZE | 4 | uint32 t | File size in bytes (BYTE)                                                    |

## 10. 3. 2 Active upload audio and video resource directory response message

Link type: slave link

Message direction: From the upper-level platform to the lower-level platform

Sub-service type identifier: DOWN\_FILELIST\_MSG\_ACK.

Description: The government video supervision platform responds to the enterprise video monitoring platform's request message for actively uploading the audio and video resource directory. The data body is shown in Table 48.

Table 48 Active upload audio and video resource directory response message

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                                                                            |
|---------------|-----------------|-------------|-------------------------------------------------------------------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                                                                                    |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006                                                            |
| DATA TYPE     | 2               | uint16t     | Sub-business type identification                                                                                        |
| DATA LENGTH   | 4               | uint32t     | Length of data for the next 2 fields                                                                                    |
| RESULT        | 1               | BYTE        | Answer results, defined as follows:<br>0x00: Success;<br>0x01: Failure;<br>0x02: Not supported;<br>0x03: End of session |
| ITEM NUMBER   | 1               | BYTE        | Total number of resource catalogs                                                                                       |

## 10. 3. 3 Query Audio/Video Resource Catalog Request Message

Sub-service type identification: DOWN a REALVIDEO a FILELIST a REQ.

DESCRIPTION: The government video monitoring platform initiates a query message for audio/video resource catalog request to the enterprise video monitoring platform, or the higher-level government platform initiates a query message for the lower-level government platform. The data body is shown in Table 49.

Table 49 Query Audio/Video Resource Catalog Request Data Bodies

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                                         |
|---------------|-----------------|-------------|--------------------------------------------------------------------------------------|
| VEHICLE_NO    | 21              | OctetString | license plate number                                                                 |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006                         |
| DATA TYPE     | 2               | uint16t     | Sub-business type identification                                                     |
| DATA_LENGTH   | 4               | uint32t     | Length of data for the next 9 fields                                                 |
| CHANNEL_ID    | 1               | BYTE        | Logical channel number, according to Table 2 in JT/T 1076-2016, 0 means all channels |



Table 49 (continued)

| Field Name     | Number of bytes | Data Type | Description and Requirements                                                                                                                                                                               |
|----------------|-----------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| START TIME     | 8               | time t    | Start time, YY-MM-DD-HH-MM-SS, all 0 means no start time condition                                                                                                                                         |
| END TIME       | 8               | time t    | Termination time, YY-MM-DD-HH-MM-SS, all 0 means no termination time condition                                                                                                                             |
| ALARM TYPE     | 8               | BYTES     | Alarm type.<br>bit0-31 See the definition of alarm flag bit in Table 18 of JT/T 808-2011;<br>bit32-63 See Table 9;<br>All 0 means no alarm type condition                                                  |
| AVITEM TYPE    | 1               | BYTE      | Audio/Video Type, 0: Audio/Video; 1: Audio; 2: Video; 3: Video or Audio/Video                                                                                                                              |
| STREAM TYPE    | 1               | BYTE      | Stream type, 0: all streams, 1: main stream, 2: sub-streams                                                                                                                                                |
| MEM TYPE       | 1               | BYTE      | Memory type, 0: all storage 1: primary storage, 2: disaster recovery storage                                                                                                                               |
| AUTHORIZE CODE | 64              | BYTES     | password (used by sentry)                                                                                                                                                                                  |
| GNSS DATA      | 36              | BYTES     | Any location within 5 minutes after the vehicle enters the cross-domain area. this field is used only for cross-domain access requests. in accordance with the protocol 4.5.8.1. in JT/T 809-2011. 5. 8. 1 |

#### 10. 3. 4 Query Audio/Video Resource Catalog Answer Message

Link type: primary link

Message direction: lower platform to higher platform.

Sub-service type identifier: UP REALVIDEO FILELIST REQ ACK.

DESCRIPTION: The enterprise video surveillance platform responds to the government video monitoring platform or the lower level government platform responds to the higher level government platform with audio and video resource catalog messages. The data body is shown in Table 50.

Table 50 Query Audio/Video Resource Catalog Response Data Bodies

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                 |
|---------------|-----------------|-------------|--------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                         |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006 |
| DATA TYPE     | 2               | uint16 t    | Sub-business type identification                             |

|             |   |          |                                      |
|-------------|---|----------|--------------------------------------|
| DATA LENGTH | 4 | uint32 t | Length of data for the next 3 fields |
|-------------|---|----------|--------------------------------------|

Table 50 (continued)

| Field Name | Number of bytes | Data Type | Description and Requirements                                                                                                                                                                           |
|------------|-----------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RESULT     | 1               | BYTE      | Answer results, defined as follows:<br>0x00: Success;<br>0x01: Failure;<br>0x02: Not supported;<br>0x03: End of session;<br>0x04: Timing password error;<br>0x05: Cross-domain condition not satisfied |
| ITEM NUM   | 4               | DWORD     | Total number of resource catalog entries                                                                                                                                                               |
| ITEM LIST  |                 |           | For a list of resource catalog entries, see Table 47                                                                                                                                                   |

#### 10.4 Remote video playback

##### 10.4.1 Remote Video Playback Request Message

Link type: slave link

Message direction: originator platform to receiver platform

Sub-service type identification: DOWN PLAYBACK MSG STARTUP.

Description: The government video monitoring platform sends the command to the enterprise video monitoring platform, the higher-level government platform to the lower-level government platform or the cross-domain regional government platform to the government platform of the belonging region to request the videotaped audio and video of the vehicle . The data body is shown in Table 51.

Table 51 Remote video playback request dataset

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                                         |
|---------------|-----------------|-------------|--------------------------------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                                                 |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006                         |
| DATA TYPE     | 2               | uint16 t    | Sub-business type identification                                                     |
| DATA LENGTH   | 4               | uint32 t    | Length of data in subsequent 8 fields                                                |
| CHANNEL ID    | 1               | BYTE        | Logical channel number, according to table 2 in JT/T 1076-2016, 0 means all channels |
| AVITEM TYPE   | 1               | BYTE        | Audio/Video Type, 0: Audio/Video; 1: Audio; 2: Video; 3: Video or Audio/Video        |

|                    |   |        |                                                                              |
|--------------------|---|--------|------------------------------------------------------------------------------|
| STREAM TYPE        | 1 | BYTE   | Stream type, 0: all streams, 1: main stream, 2: sub-streams                  |
| MEM TYPE           | 1 | BYTE   | Memory type, 0: all storage 1: primary storage, 2: disaster recovery storage |
| PLAYBACK STARTTIME | 8 | time t | Playback start time, UTC time                                                |

Table 51 (continued)

| Field Name       | Number of bytes | Data Type | Description and Requirements                                                                                                                                                     |
|------------------|-----------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PLAYBACK ENDTIME | 8               | time t    | Playback end time, UTC time                                                                                                                                                      |
| AUTHORIZE CODE   | 64              | BYTES     | password (used by sentry)                                                                                                                                                        |
| GNSS DATA        | 36              | BYTES     | Any location within 5 minutes after the vehicle enters the cross-domain area. this field is used only for cross-domain access requests. in accordance with JT/T 809-2011 4.5.8.1 |

## 10.4.2 Remote Video Playback Request Answer Message

Link type: primary link

Message direction: receiver platform to originator platform

Sub-service type identification: UP PLAYBACK MSG STARTUP ACK.

Description: The enterprise video surveillance platform responds to the video playback request messages sent by the government video monitoring platform, the lower-level government platform responds to the higher-level government platform, or the government platform of the belonging region responds to the government platform of the cross-domain region. The data body is shown in Table 52.

Table 52 Remote Video Playback Answer Dataset

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                                                                                                                                                           |
|---------------|-----------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                                                                                                                                                                   |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006                                                                                                                                           |
| DATA TYPE     | 2               | uint16 t    | Sub-business type identification                                                                                                                                                                       |
| DATA LENGTH   | 4               | uint32 t    | Length of data for the next 3 fields                                                                                                                                                                   |
| SERVER IP     | 32              | OctetString | Enterprise Video Server IP Address                                                                                                                                                                     |
| SERVER PORT   | 2               | uint16 t    | Enterprise Video Server Side Slogan                                                                                                                                                                    |
| RESULT        | 1               | BYTE        | Answer results, defined as follows:<br>0x00: Success;<br>0x01: Failure;<br>0x02: Not supported;<br>0x03: End of session;<br>0x04: Timing password error;<br>0x05: Cross-domain condition not satisfied |

## 10.4.3 Remote video playback control messages

Link type: slave link

Message Direction: Government Video Supervision Platform to Enterprise Video Surveillance Platform

Sub-service type identification: DOWN PLAYBACK MSG CONTROL

DESCRIPTION: The government video supervision platform sends this command to the enterprise video surveillance platform to control the playback . The data body is shown in Table 53.

Table 53 Remote video playback control dataset

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                                                                                                                                                                                                 |
|---------------|-----------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                                                                                                                                                                                                         |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006                                                                                                                                                                                 |
| DATA TYPE     | 2               | uint16 t    | Sub-business type identification                                                                                                                                                                                                             |
| DATA LENGTH   | 4               | uint32 t    | Length of data for the next 3 fields                                                                                                                                                                                                         |
| CONTROL TYPE  | 1               | BYTE        | 0x00: normal playback; 0x01: pause playback; 0x02: end playback; 0x03: fast-forward playback; 0x04: keyframe snapback playback; 0x05: drag playback; 0x06: Keyframe playback                                                                 |
| FAST TIME     | 1               | BYTE        | Fast-forward or fast-reverse multiplier, the content of this field is valid when the playback control is 0x03 and 0x04, otherwise it is set to zero.<br>0x00: Invalid;<br>0x01: 1x;<br>0x02: 2x;<br>0x03: 4x;<br>0x04: 8x;<br>0x05: 16 times |
| DATE TIME     | 8               | time t      | Drag position in UTC time, valid if playback control is 0x05                                                                                                                                                                                 |

## 10.4.4 Remote Video Playback Control Answer Message

Link type: primary link

Message direction: enterprise video surveillance platform to the government video monitoring platform

Sub-service type identification: UP PLAYBACK MSG CONTROL ACK.

DESCRIPTION: The enterprise video surveillance platform responds to the playback control messages sent by the government video supervision platform . See Table 54 for data bodies.

Table 54 Remote Video Playback Answer Data Bodies

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                 |
|---------------|-----------------|-------------|--------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                         |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006 |

|             |   |          |                                                                                                                         |
|-------------|---|----------|-------------------------------------------------------------------------------------------------------------------------|
| DATA TYPE   | 2 | uint16 t | Sub-business type identification                                                                                        |
| DATA LENGTH | 4 | uint32 t | Subsequent 1 field data length                                                                                          |
| RESULT      | 1 | BYTE     | Answer results, defined as follows:<br>0x00: Success;<br>0x01: Failure;<br>0x02: Not supported;<br>0x03: End of session |



## 10.5 Remote video download

## 10.5.1 Remote Video Download Request Message

Link type: slave link

Message Direction: Government Video Supervision Platform to Enterprise Video Surveillance Platform

Sub-service type identification: DOWN a DOWNLOAD a MSG a STARTUP.

Description: The government video monitoring platform sends this command to the enterprise video monitoring platform to download the video recording audio and video of the vehicles. The data body is shown in Table 55.

Table 55 Remote Video Download Request Data Body

| Field Name     | Number of bytes | Data Type   | Description and Requirements                                                                                                                                                            |
|----------------|-----------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VEHICLE NO     | 21              | OctetString | license plate number                                                                                                                                                                    |
| VEHICLE COLOR  | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006                                                                                                                            |
| DATA_TYPE      | 2               | uint16_t    | Sub-business type identification                                                                                                                                                        |
| DATA_LENGTH    | 4               | uint32_t    | Subsequent 10 field data lengths                                                                                                                                                        |
| CHANNEL_ID     | 1               | BYTE        | Logical channel number, according to table 2 in JT/T 1076-2016, 0 means all channels                                                                                                    |
| START_TIME     | 8               | time_t      | UTC time                                                                                                                                                                                |
| END_TIME       | 8               | time_t      | UTC time                                                                                                                                                                                |
| ALARM_TYPE     | 8               | 64BITS      | bit0 to bit31 See Table 18 of JT/T 808-2011 for the definition of alarm flag bits;<br>See Table 10 for bit32 to bit63.                                                                  |
| AVITEM_TYPE    | 1               | BYTE        | Audio and video types, defined as follows:<br>0x00: audio and video; 0x01: audio; 0x02: video                                                                                           |
| STREAM TYPE    | 1               | BYTE        | The code stream type, defined as follows:<br>0x01: main stream; 0x02: sub-streams                                                                                                       |
| MEM TYPE       | 1               | BYTE        | memory type, defined as follows:<br>0x01: main memory; 0x02: disaster memory                                                                                                            |
| FILE SIZE      | 4               | uint32_t    | File size in bytes (BYTE)                                                                                                                                                               |
| AUTHORIZE CODE | 64              | BYTES       | password (used by sentry)                                                                                                                                                               |
| GNSS DATA      | 36              | BYTES       | Any position within 5min after the vehicle enters the cross-domain area. this field is used only for cross-domain access requests. see protocol 4 in JT/T 809-2011 for details. 5. 8. 1 |

#### 10.5.2 Remote Video Download Request Answer Message

Link type: primary link

Message direction: enterprise video surveillance platform to the government video monitoring platform

Sub-service type identification: DOWN-DOWNLOAD-MSG-STARTUP.

Description: Answer message sent by the enterprise video surveillance platform to the government video monitoring platform for downloading vehicle audio and video requests. The data body is shown in Table 56.

Table 56 Remote Video Download Request Response Dataset

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                                                                                                                                                           |
|---------------|-----------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VEHICLE-NO    | 21              | OctetString | license plate number                                                                                                                                                                                   |
| VEHICLE-COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006                                                                                                                                           |
| DATA-TYPE     | 2               | uint16-t    | Sub-business type identification                                                                                                                                                                       |
| DATA-LENGTH   | 4               | uint32-t    | Length of data for the next 2 fields                                                                                                                                                                   |
| RESULT        | 1               | BYTE        | Answer results, defined as follows:<br>0x00: Success;<br>0x01: Failure;<br>0x02: Not supported;<br>0x03: End of session;<br>0x04: Timing password error;<br>0x05: Cross-domain condition not satisfied |
| SESSION-ID    | 2               | uint16-t    | Corresponds to the flow number of the platform file upload message, valid when RE-SULT is 0                                                                                                            |

### 10.5.3 Remote video download completion notification message

Link type: primary link

Message direction: enterprise video surveillance platform to the government video monitoring platform

Sub-service type identification: UP-DOWNLOAD-MSG-END-INFORM.

Description: The enterprise video surveillance platform sends to the government video supervision platform to notify the government video supervision platform that the video recording file has been downloaded from the terminal Completed . The data body is shown in Table 57.

Table 57 Remote video download completion notification dataset

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                 |
|---------------|-----------------|-------------|--------------------------------------------------------------|
| VEHICLE-NO    | 21              | OctetString | license plate number                                         |
| VEHICLE-COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006 |
| DATA-TYPE     | 2               | uint16-t    | Sub-business type identification                             |
| DATA-LENGTH   | 4               | uint32-t    | Length of data for the next 7 fields                         |

|            |    |             |                                                                                             |
|------------|----|-------------|---------------------------------------------------------------------------------------------|
| RESULT     | 1  | BYTE        | 0x00: Success.<br>0x01: Failure                                                             |
| SESSION-ID | 2  | uint16-t    | Corresponds to the flow number of the platform file upload message, valid when RE-SULT is 0 |
| SERVER-IP  | 32 | OctetString | FTP server ip address, valid if RESULT is 0                                                 |

Table 57 (continued)

| Field Name | Number of bytes | Data Type   | Description and Requirements              |
|------------|-----------------|-------------|-------------------------------------------|
| TCP PORT   | 2               | uint16 t    | FTP port, valid when RESULT is 0          |
| USERNAEM   | 49              | OctetString | FTP user name, valid if RESULT is 0       |
| PASSWORD   | 22              | OctetString | FTP password, valid if RESULT is 0        |
| FILE PATH  | 200             | OctetString | File storage path, valid when RESULT is 0 |

## 10.5.4 Remote Video Download Completion Notification Answer Message

Link type: slave link

Message Direction: Government Video Supervision Platform to Enterprise Video Surveillance Platform

Sub-service type identification: UP DOWNLOAD MSG END INFORM ACK.

Description: Government video monitoring platform responds to enterprise video surveillance platform download completion notification . See Table 58 for the data body.

Table 58 Remote Video Download Completion Notification Response Dataset

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                                                        |
|---------------|-----------------|-------------|-----------------------------------------------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                                                                |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006                                        |
| DATA TYPE     | 2               | uint16 t    | Sub-business type identification                                                                    |
| DATA LENGTH   | 4               | uint32 t    | Length of data for the next 2 fields                                                                |
| RESULT        | 1               | BYTE        | Answer results:<br>0x00: Success;<br>0x01: Failure;<br>0x02: Not supported;<br>0x03: End of session |
| SESSION ID    | 2               | uint16 t    | Corresponds to the flow number of the platform file upload message, valid when RE-SULT is 0         |

## 10.5.5 Remote Video Download Control Request Message

Link type: slave link

Message Direction: Government Video Supervision Platform to Enterprise Video Surveillance Platform

Sub-service type identification: DOWN DOWNLOAD MSG CONTROL.

DESCRIPTION: The government video supervision platform sends a download control message to the enterprise video surveillance platform . The data body is shown in Table 59.

Table 59 Remote video download control request dataset

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                 |
|---------------|-----------------|-------------|--------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                         |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006 |

Table 59 (continued)

| Field Name  | Number of bytes | Data Type | Description and Requirements                                           |
|-------------|-----------------|-----------|------------------------------------------------------------------------|
| DATA TYPE   | 2               | uint16 t  | Sub-business type identification                                       |
| DATA LENGTH | 4               | uint32 t  | Length of data for the next 2 fields                                   |
| SESSION ID  | 2               | uint16 t  | Corresponds to the stream number of the platform's file upload message |
| TYPE        | 1               | BYTE      | 0x00: Pause;<br>0x01: Continue;<br>0x02: Cancel                        |

## 10.5.6 Remote Video Download Control Request Answer Message

Link type: primary link

Message direction: enterprise video surveillance platform to the government video monitoring platform

Sub-service type identification: UP DOWNLOAD MSG CONTROL ACK.

DESCRIPTION: Answer message of the enterprise video surveillance platform to the download control request sent by the government video supervision platform . The data body is shown in Table 60.

Table 60 Remote video download control request response dataset

| Field Name    | Number of bytes | Data Type   | Description and Requirements                                                                        |
|---------------|-----------------|-------------|-----------------------------------------------------------------------------------------------------|
| VEHICLE NO    | 21              | OctetString | license plate number                                                                                |
| VEHICLE COLOR | 1               | BYTE        | Color of license plate, according to 5.4.12 of JT/T 415-2006                                        |
| DATA TYPE     | 2               | uint16 t    | Sub-business type identification                                                                    |
| DATA LENGTH   | 4               | uint32 t    | Subsequent 1 field data length                                                                      |
| RESULT        | 1               | BYTE        | Answer results:<br>0x00: Success;<br>0x01: Failure;<br>0x02: Not supported;<br>0x03: End of session |

## Appendix A

(Normative appendix)

Comparison table of messages between video terminals and video platforms

See Table A. 1 for a message cross-reference table of communication protocols between video terminals and video platforms.

Table A.1 Comparison of messages between video terminals and video platforms

| serial number | Message Body Name                                                     | Message ID | serial number | Message Body Name                               | Message ID |
|---------------|-----------------------------------------------------------------------|------------|---------------|-------------------------------------------------|------------|
| 1             | Query terminal audio/video attributes                                 | 0x9003     | 12            | File Upload Command                             | 0x9206     |
| 2             | Terminal upload audio and video attributes                            | 0x1003     | 13            | Notification of file upload completion          | 0x1206     |
| 3             | Real-time audio and video transmission requests                       | 0x9101     | 14            | File Upload Control                             | 0x9207     |
| 4             | Terminal uploads passenger traffic                                    | 0x1005     | 15            | PTZ                                             | 0x9301     |
| 5             | Audio and video real-time transmission control                        | 0x9102     | 16            | Gimbal Adjustment Focus Control                 | 0x9302     |
| 6             | Real-time audio and video streaming and passthrough data transmission |            | 17            | Gimbal Adjustment Aperture Control              | 0x9303     |
| 7             | Real-time audio and video transmission status notification            | 0x9105     | 8             | Head Wiper Control                              | 0x9304     |
| 8             | Query Resource List                                                   | 0x9205     | 9             | Infrared fill light control                     | 0x9305     |
| 9             | List of audio and video resources uploaded by terminals               | 0x1205     | 20            | Gimbal zoom control                             | 0x9306     |
| 10            | Remote video playback requests from the Platform                      | 0x9201     | 21            | Platform manual wake-up request (short message) | WAKEUPXX   |
| 11            | Platform downlink remote video playback control                       | 0x9202     |               |                                                 |            |